

Department of Computer Science and Engineering (CSE)

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Course: Machine Learning (CSE475)

Project Name: Rice varieties (microscopic)

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1. Dataset Overview

The exploratory data analysis was performed on the PRBD Microscopic Image of Different Processed Rice dataset. The dataset contains 10,000 images belonging to 10 rice classes. Among these, 2,000 are original images and 8,000 are augmented images. Each rice class contains 1,000 images in total, providing an equal overall class distribution.

The metadata includes image paths, filenames, class labels, image types, source image IDs, file extensions, and file sizes. This structured information was used to examine the dataset's organization and quality.

2. Rice Class Distribution

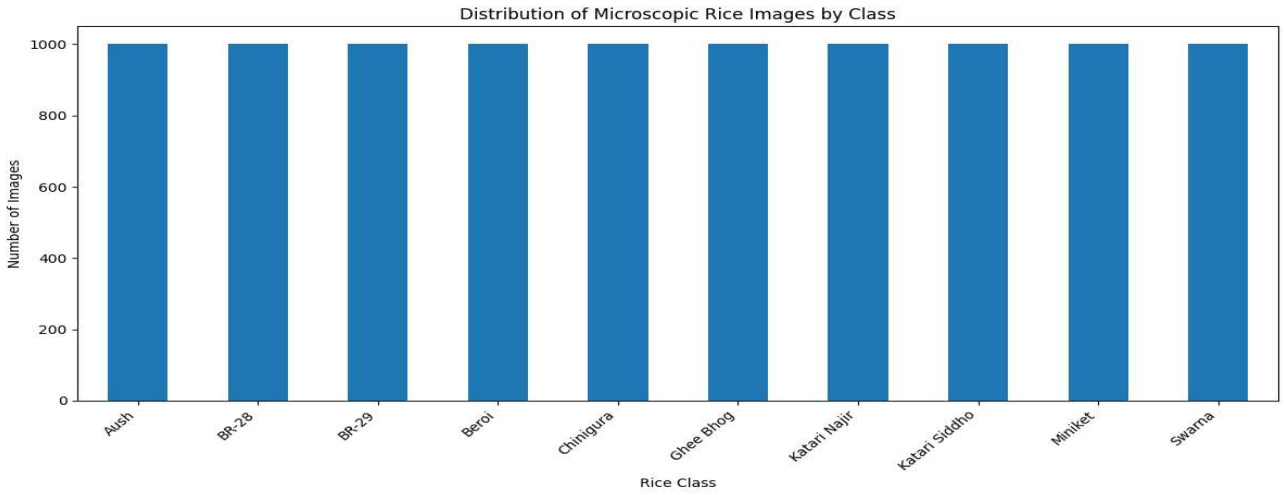


Figure 2.1: Distribution of Microscopic Rice Images by Class

The class distribution analysis shows that all ten rice classes contain exactly 1,000 images each. The classes are Aush, BR-28, BR-29, Beroi, Chinigura, Ghee Bhog, Katari Najir, Katari Siddho, Miniket, and Swarna.

Finding: The overall dataset has a balanced class distribution, with each class representing 10% of the total images. This provides equal image representation across the ten categories. However, this balance includes augmented images and does not mean that all 10,000 images are independent original samples.

3. Original and Augmented Image Distribution

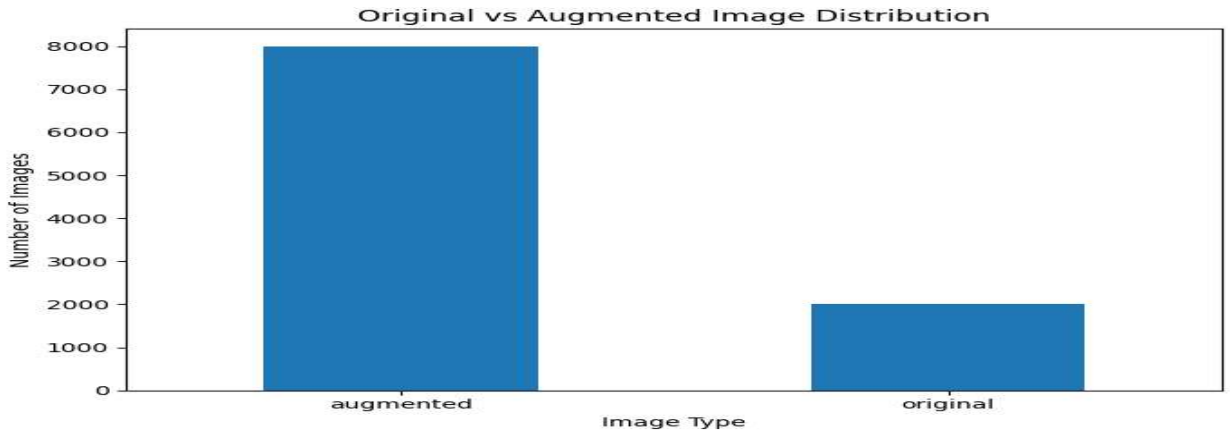


Figure 3.1: Original vs Augmented Image Distribution

The analysis identified 2,000 original images and 8,000 augmented images. Therefore, original images account for 20% of the dataset, while augmented images account for 80%.

Finding: Data augmentation increases the number of available images by creating transformed versions of original samples. However, the high proportion of augmented images means that model evaluation must account for similarities between images derived from the same source.

4. Class-wise Original and Augmented Images

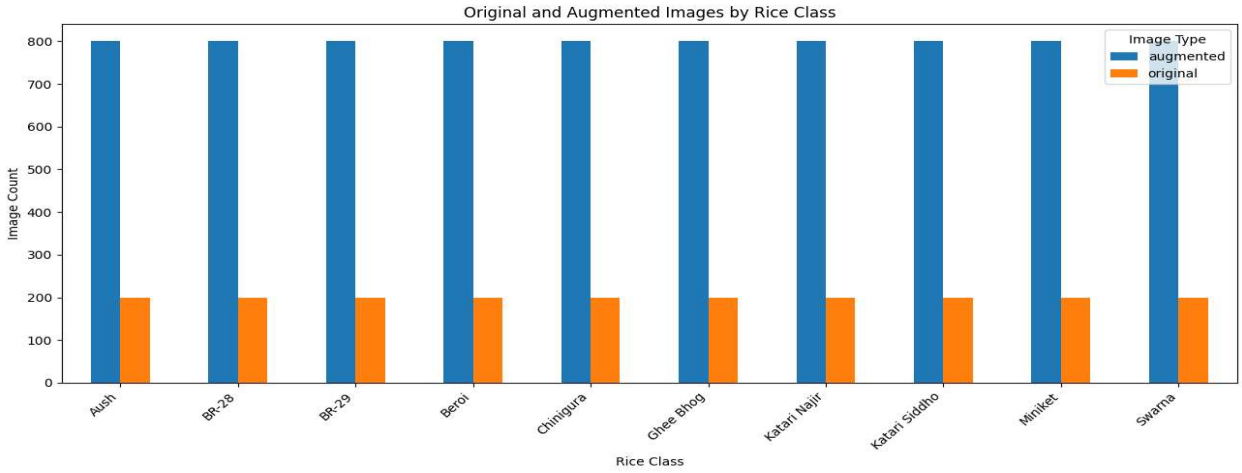


Figure 4.1: Original and Augmented Images by Rice Class

The class-wise analysis shows that each rice class contains 200 original images and 800 augmented images. Thus, the original-to-augmented distribution is consistent across all ten classes.

Finding: Augmentation has been applied equally across all rice categories. Each class has the same number of original and augmented images, contributing to consistent class representation throughout the dataset.

5. Visual Inspection of Microscopic Rice Images

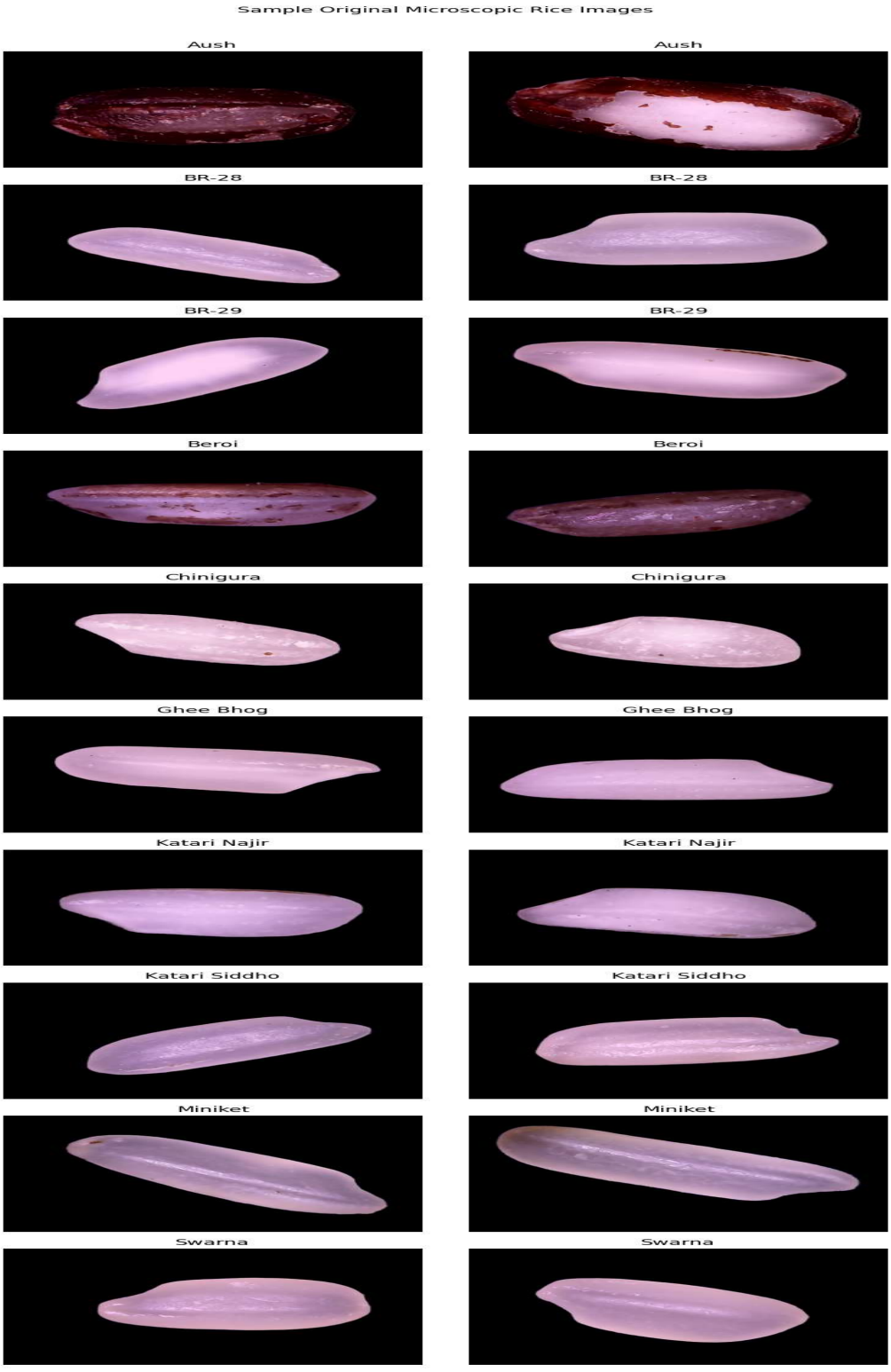


Figure 5.1: Sample Original Microscopic Rice Images

The visual inspection presents two sample original images from each rice class. The images show differences in grain shape, size, surface texture, color, and visible structural characteristics.

Finding: The samples demonstrate visual variation among the rice categories. These characteristics may provide useful features for image classification. However, visual inspection alone cannot establish how accurately a machine learning model can distinguish the classes.

6. Image Dimensions and File Size Analysis

The following results were obtained:

Image property	Observed result
Images inspected	500
Mean image width	640 pixels
Mean image height	480 pixels
Minimum file size	15.48 KB
Mean file size	43.57 KB
Median file size	25.75 KB
Maximum file size	226.42 KB
RGB images	406 (81.2%)
RGBA images	94 (18.8%)

Finding: All 500 inspected images have identical dimensions of 640 × 480 pixels. However, their file sizes vary considerably, ranging from approximately 15.48 KB to 226.42 KB. Most inspected images use RGB color mode, while the remaining images use RGBA, which includes an alpha channel.

These results suggest consistent spatial dimensions in the inspected sample, although color-mode consistency may need to be addressed during preprocessing.

7. Image Width vs Height

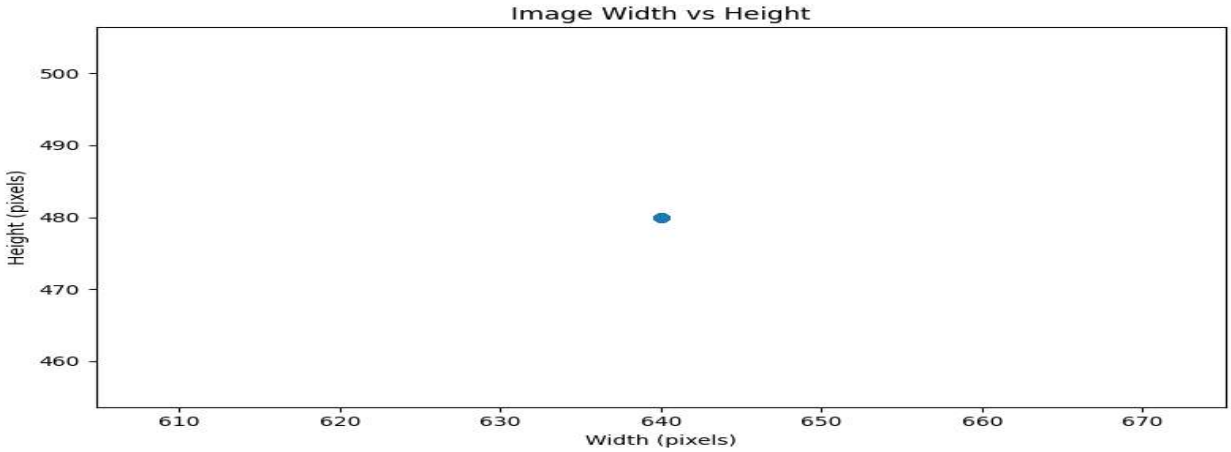


Figure 3.5: Image Width vs Height

The scatter plot displays image width against image height for the 500 inspected images. All points overlap at 640 pixels in width and 480 pixels in height.

Finding: No variation in image dimensions was observed in the inspected sample. Therefore, resizing may not be necessary to standardize image dimensions for this sample, although the remaining images should also be checked before making a conclusion about the entire dataset.

8. Distribution of Image File Sizes

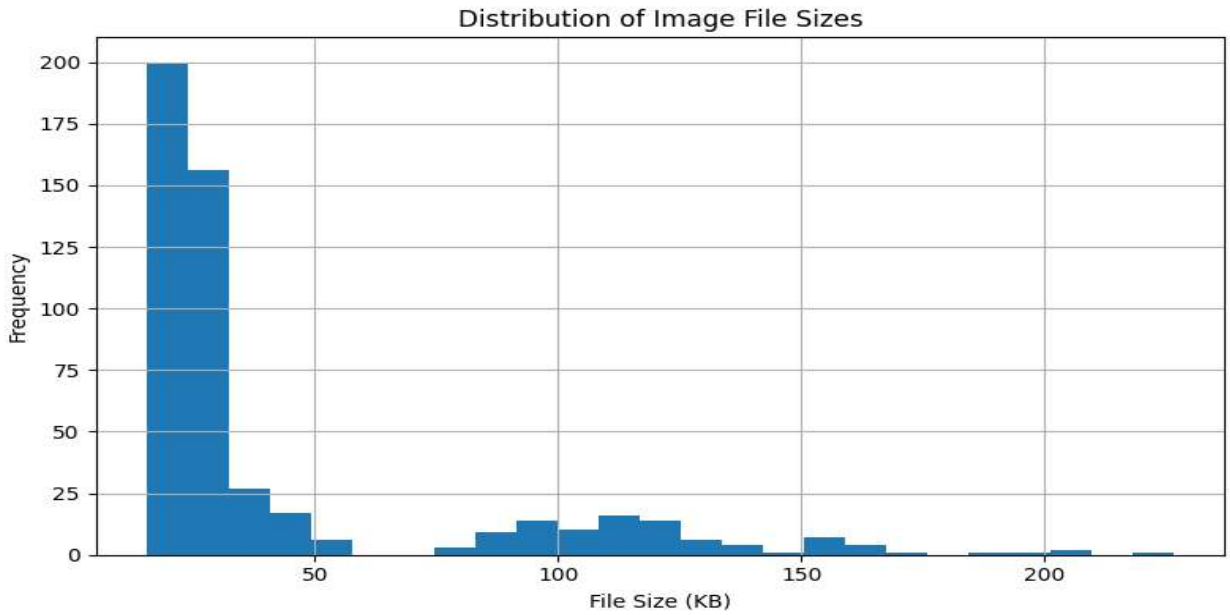


Figure 3.6: Distribution of Image File Sizes

The histogram illustrates the frequency distribution of file sizes among the 500 inspected images. The distribution is concentrated toward smaller file sizes, with fewer images occupying substantially more storage.

Finding: Image file sizes vary despite identical dimensions. This indicates differences in image content, encoding, or compression. The observed maximum of 226.42 KB is considerably higher than the median of 25.75 KB. These variations are relevant to storage requirements and data-loading efficiency.

9. Class Imbalance Analysis

The class imbalance analysis was conducted using original images only.

Measurement	Result
Maximum original images in a class	200
Minimum original images in a class	200
Imbalance ratio	1.0

Finding: All ten classes contain exactly 200 original images. The imbalance ratio of 1.0 indicates that the original dataset is perfectly balanced across the rice classes. Therefore, no class has a numerical advantage over another in terms of original-image count.

10. Missing Values and Duplicate Checks

The metadata quality checks produced the following results:

Quality check	Result
Total metadata records	10,000
Missing values	0
Duplicate image paths	0
Duplicate filenames	4,565
Unknown class labels	0
Unknown image types	0
Non-positive file sizes	0

Finding: The metadata contains no missing values, duplicate image paths, unknown labels, unknown image types, or non-positive file sizes. However, 4,565 filenames are duplicated. This requires careful interpretation because filenames may repeat across different folders or image types; it does not automatically mean that 4,565 image files contain duplicate image data.

11. Corrupted or Unreadable Image Detection

Checked all 10,000 image files for readability.

Finding: No corrupted or unreadable images were detected. All 10,000 files passed the notebook's image verification check. This indicates that no file-readability problems were identified during this check, although it does not guarantee that every image is visually correct or correctly labelled.

12. Summary of EDA Findings

The exploratory analysis established that the PRBD dataset contains 10,000 microscopic rice images across ten classes, with 2,000 original images and 8,000 augmented images. All classes have equal representation in both the complete dataset and the original-image subset.

The inspected images consistently measure 640 × 480 pixels, but their file sizes and color modes vary. Metadata validation identified no missing values or unreadable images, although repeated filenames require further examination.

Overall, the dataset demonstrates balanced class representation and consistent image dimensions within the inspected sample. Before model training, attention should be given to color-mode preprocessing, duplicate verification, and source-based splitting of original and augmented images to prevent data leakage.

Important limitation: The EDA notebook does not train or evaluate a classification model. Therefore, these findings describe the dataset's characteristics and quality; they do not establish classification accuracy or predictive performance.